

Predicting Hospital Inpatient Charges

New York State SPARCS 2021 · 50,000 Records · 7 Models · $R^2 = 0.814$

Boston University · BA810 Supervised Machine Learning · Fall 2025

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THE PROBLEM

Hospital Charge Variation Exceeds 10× Across New York State

The core question

Can patient demographics, clinical characteristics, and hospital-level features predict inpatient charges accurately enough to support benchmarking, anomaly detection, and financial planning?

Scope note: This model predicts billed charges — not actual costs or payments. Predictions should be read as billing benchmarks, not cost estimates.

10×

charge variation, same diagnosis

\$73,548

average billed per stay

Hospitals

Benchmark charges against statewide norms. Flag billing outliers for revenue integrity review.

Insurers

Calibrate reimbursement rates and support actuarial modeling.

Policymakers

Monitor regional and demographic pricing disparities across facilities.

THE DATA

SPARCS 2021: 50,000 NY State Inpatient Discharge Records, 33 Variables

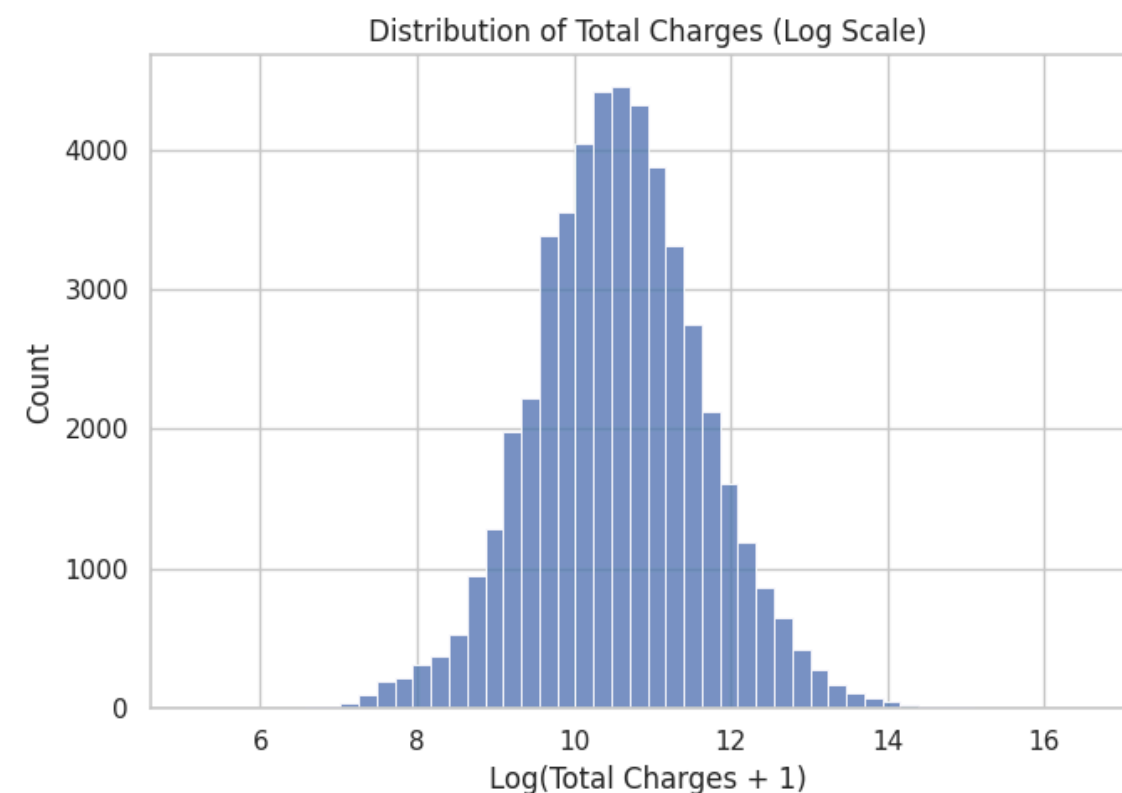
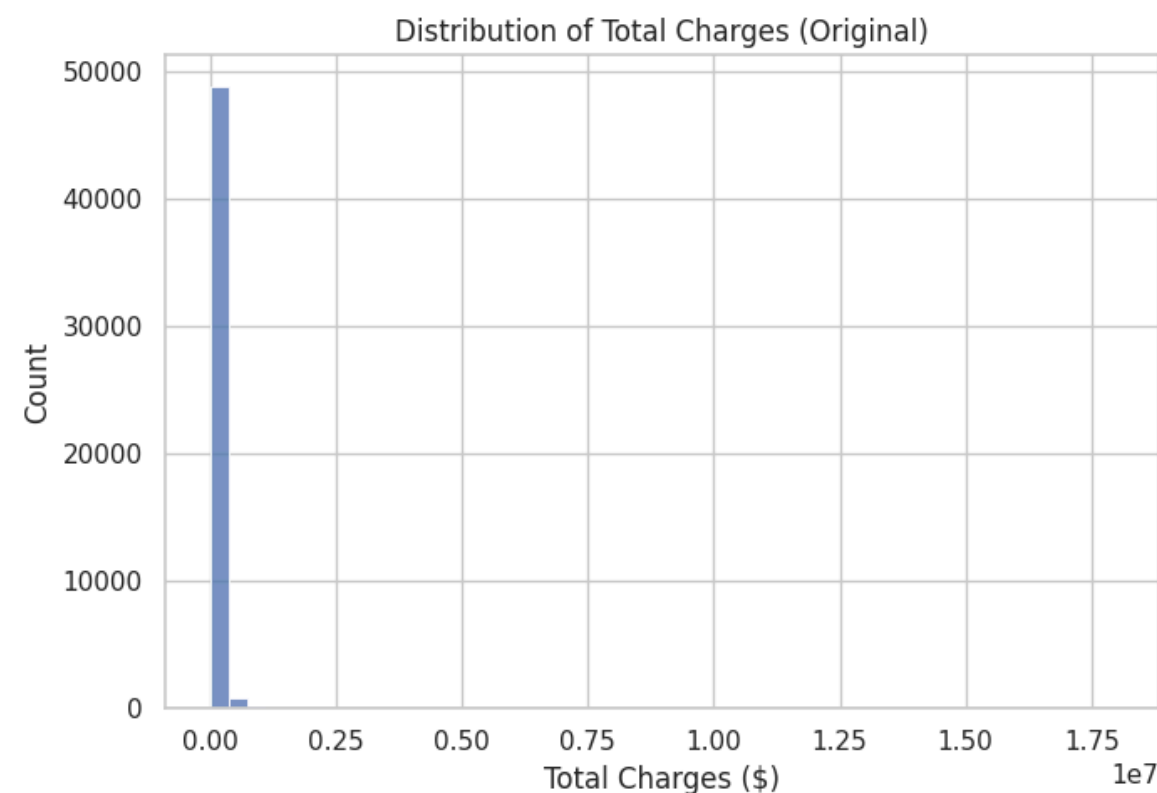
50,000
discharge records

33
features per record

2021
SPARCS statewide census

Category	Key Fields
Hospital & Facility	hospital_county, hospital_service_area, facility_name
Patient Demographics	age_group, gender, race, ethnicity, zip_code_3_digits
Visit Characteristics	length_of_stay, type_of_admission, patient_disposition
Clinical Classification	apr_drg_code, apr_severity_of_illness_code, apr_risk_of_mortality
Target	total_charges (total_costs excluded — target leakage risk)

Charges Are Severely Right-Skewed — Log Transformation Required Before Modeling

**36.07**

skewness (raw charges)

\$38,376

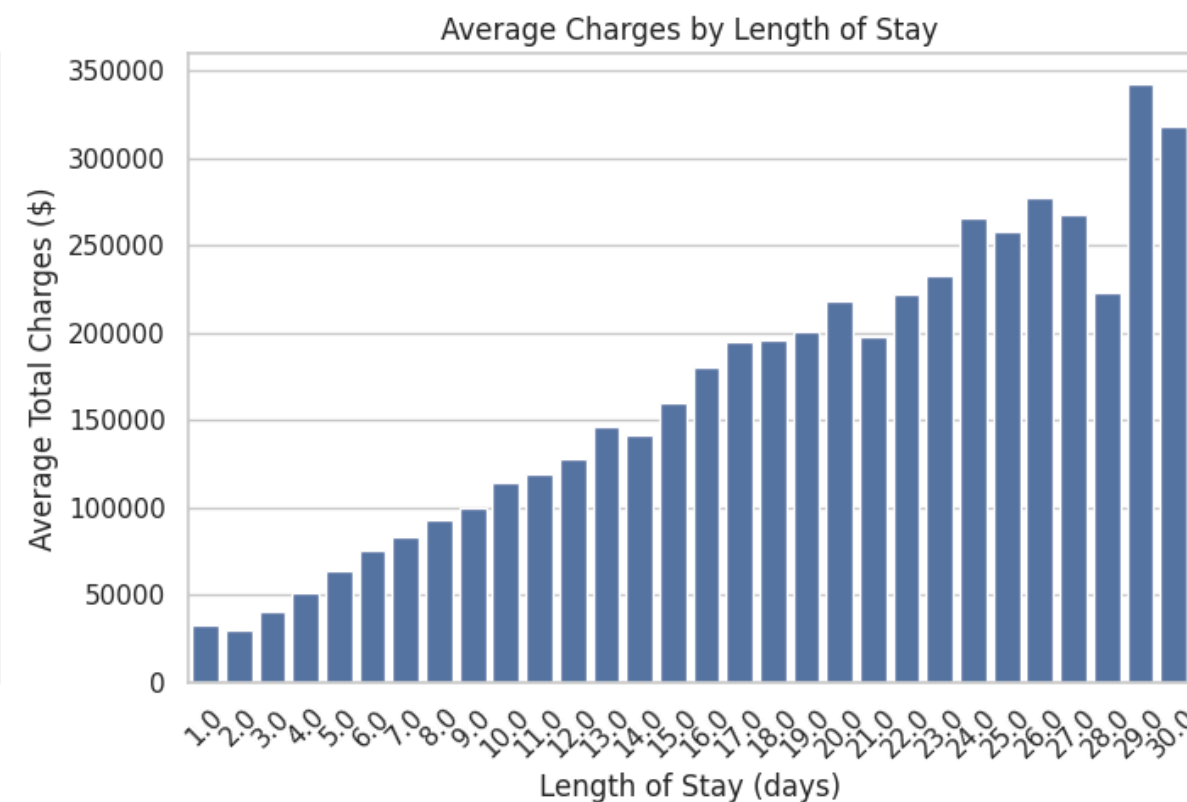
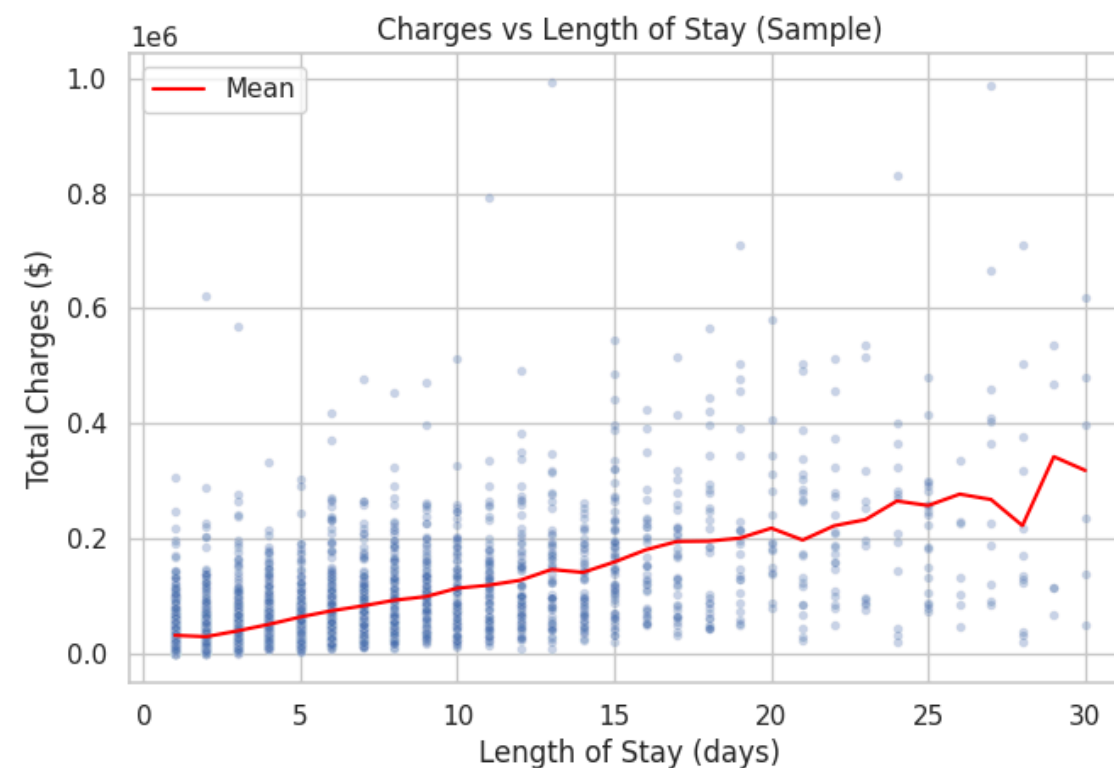
median charge

\$179,780

standard deviation

Preprocessing decision: To prevent extreme cases from dominating the loss function while preserving full-dataset evaluation in dollar terms, we log-transform the target using `TransformedTargetRegressor(np.log1p/np.expm1)` and Winsorize it at the 99th percentile (\$556,254).

Length of Stay Is the Single Strongest Predictor — Correlation $r = 0.677$



$r = 0.677$

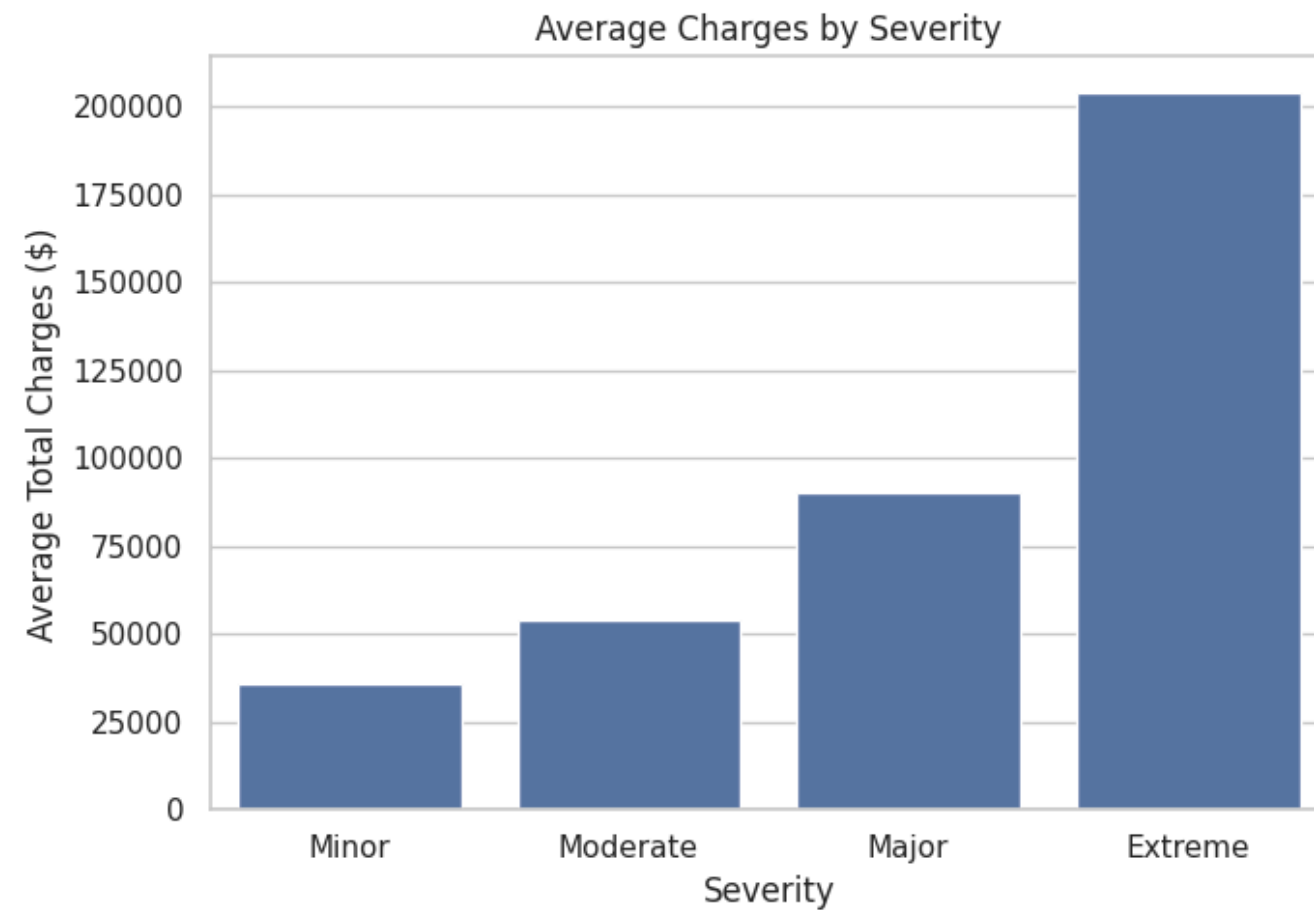
LOS × charge correlation

Key observations

- Every additional day → more services, nursing care, and billing
- The relationship is nonlinear — steeper at shorter stays
- LOS is a visit characteristic, not knowable at admission
- Its inclusion is appropriate for retrospective benchmarking

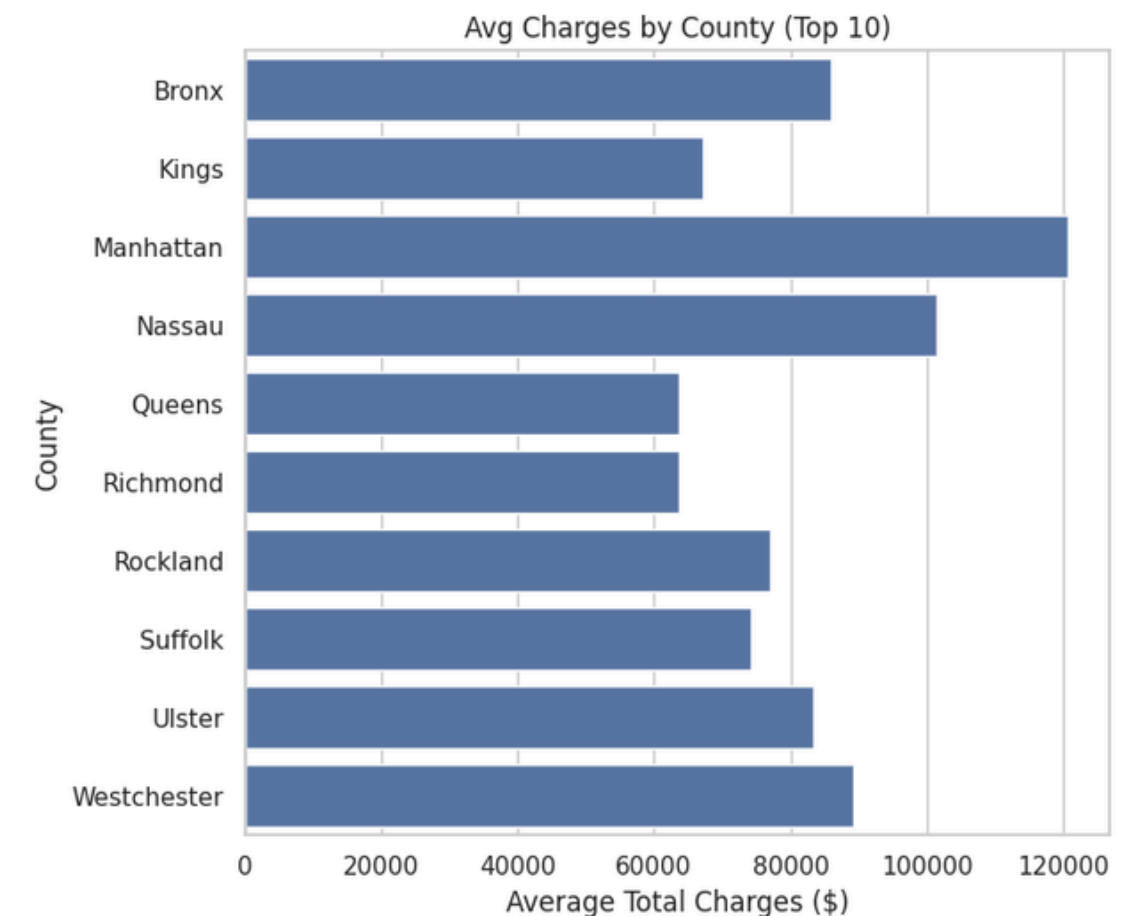
Confirmed by SHAP: length_of_stay has the largest mean |SHAP| value in the final XGBoost model — its dominance holds across both linear and nonlinear methods.

APR Severity and Hospital County Are Independent Drivers of Charge Variation



Severity drives 3–5× charge difference

Minor vs Extreme cases: avg \$32K vs \$197K. Encoded as ordinal (not nominal) to preserve rank ordering in the pipeline.



Geography adds signal beyond clinical factors

Manhattan and Nassau charge significantly more than upstate counties — facility pricing strategy, not just patient mix. Encoded via FrequencyEncoder.

A Differentiated Pipeline — Feature Type Determines Encoding Strategy

Feature Type	Treatment	Rationale
length_of_stay (numeric)	SimpleImputer — median	Handles 30 missing values from the 120+ cap
Severity & mortality codes (ordinal)	OrdinalEncoder	Preserves Minor < Moderate < Major < Extreme rank order
DRG, diagnosis, procedure codes (high-cardinality)	Custom FrequencyEncoder	100s of unique values — OHE would create sparse matrix; target encoding risks leakage
Demographics, admission type (low-cardinality)	OneHotEncoder	Standard treatment for nominal categoricals with few levels
total_charges (target)	TransformedTargetRegressor (log1p / expm1) + Winsorize 99th pctile	Right-skewed distribution; extreme values cap at \$556,254

All transformations fit on training data only — applied to test set via ColumnTransformer, preventing data leakage.

Linear Models Plateau at $R^2=0.606$ — The Limiting Factor Is Model Form, Not Regularization

Linear Regression

0.606

R^2

MAE: \$33,061

Simple baseline. Sensitive to outliers and skew.

Ridge (L2)

0.606

R^2

MAE: \$33,061

Best $\alpha=10$. No improvement — multicollinearity is not the bottleneck.

Lasso (L1)

0.606

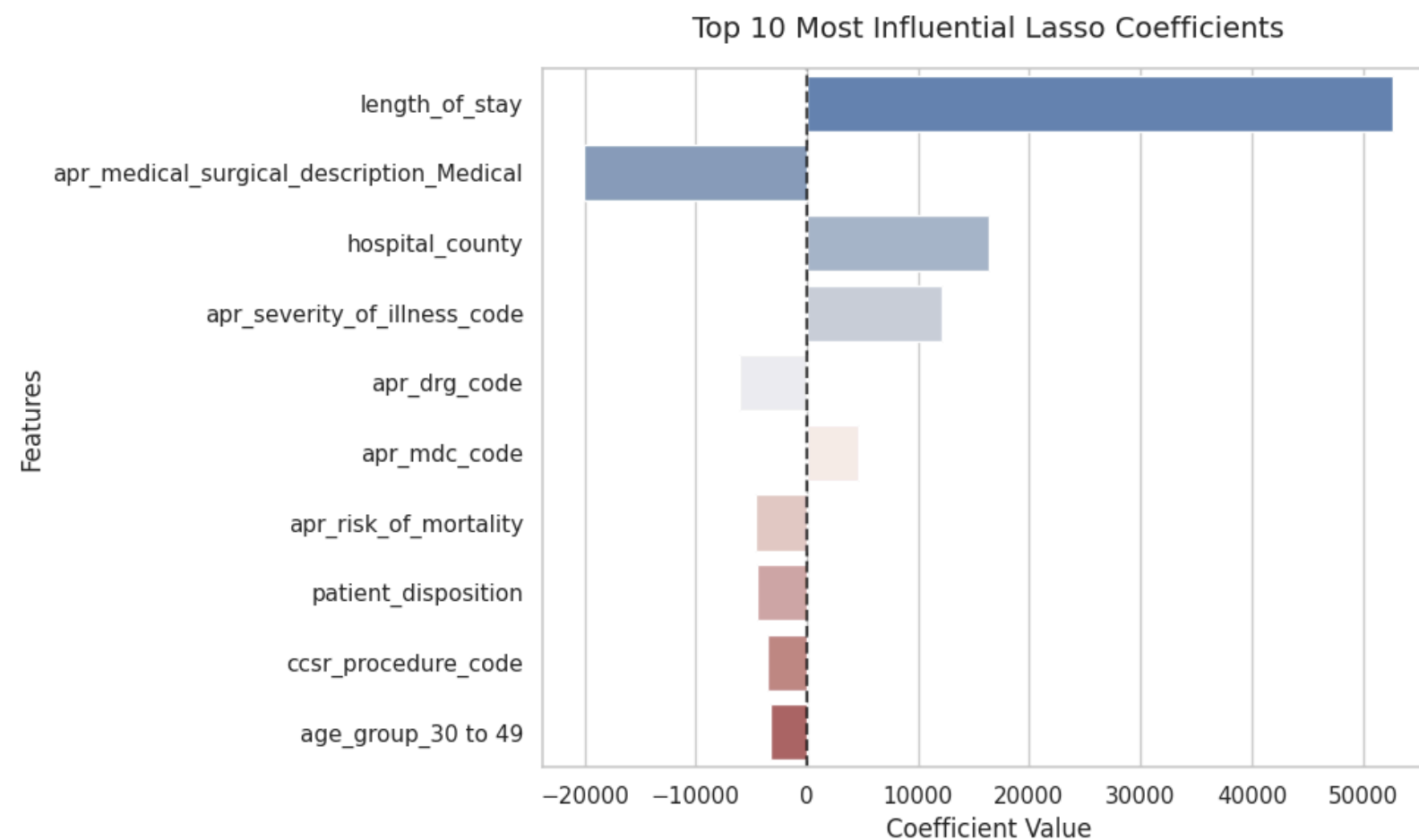
R^2

MAE: \$32,985

Best $\alpha=100$. Marginal MAE gain. Selects 34 of 46 features.

Key insight: When Ridge doesn't improve over Linear Regression, multicollinearity is not the primary problem. Hospital charge variation has meaningful nonlinear structure — regularization cannot solve a model-form limitation. Tree-based methods are needed.

Lasso Identifies the Key Cost Drivers: LOS Dominates, Then Procedure Type, Geography, and Severity



#1 length_of_stay

Coeff: 52,758 — dominates by ~2.5× the next feature

#2 Medical vs Surgical

Surgical stays billed substantially higher

#3 hospital_county

Geography adds signal beyond clinical factors

#4 APR severity code

Ordinal encoding preserves the rank ordering

These linear findings are validated and extended by SHAP analysis on the final XGBoost model — see Slide 13.

Tree Models Confirm Nonlinearity — Each Step in the Progression Answers a Question

Model	R ²	MAE	What it adds
Decision Tree	0.684	\$24,555	Proves nonlinearity exists — interaction effects captured
Random Forest	0.773	\$20,010	Bagging reduces variance; 18% MAE improvement over single tree
XGBoost (default)	0.801	\$18,675	Sequential error correction focuses on hardest cases
XGBoost (tuned)	0.814	\$18,011	Best: 31% RMSE reduction vs linear baseline

The R²=0.606 → 0.684 jump from linear to Decision Tree is the most informative result in the comparison. It confirms the data has structure that linear models cannot represent.

Tuning XGBoost gained 3.4% additional RMSE reduction (modest). Cross-validated RMSE (\$39,116) close to test RMSE (\$37,166) confirms no significant overfitting.

MODEL RESULTS

Tuned XGBoost: $R^2=0.814$, MAE=\$18,011, 31% RMSE Reduction vs. Linear Baseline

0.814

R^2 — variance explained

\$18,011

MAE — avg absolute error

31%

RMSE reduction vs. baseline

Model	MAE	RMSE	R^2
Linear / Ridge / Lasso	~\$33,000	~\$54,090	0.606
Decision Tree	\$24,555	\$48,421	0.684
Random Forest	\$20,010	\$41,047	0.773
XGBoost (default)	\$18,675	\$38,455	0.801
XGBoost (tuned)	\$18,011	\$37,166	0.814

The ~19% residual unexplained variance reflects unobserved pricing factors — hospital contract rates and charge master pricing — that are not present in administrative discharge data.

Preprocessing Design Drove as Much Performance Gain as Model Selection

Log Transform + Winsorization

✗ Without: Linear models would see MAE >\$60K; extreme outliers dominate the loss function

✓ With: Stable gradients; all models train on a learnable target distribution

OrdinalEncoder for Severity

✗ Without: OneHotEncoder treats severity levels as independent categories — loses rank signal

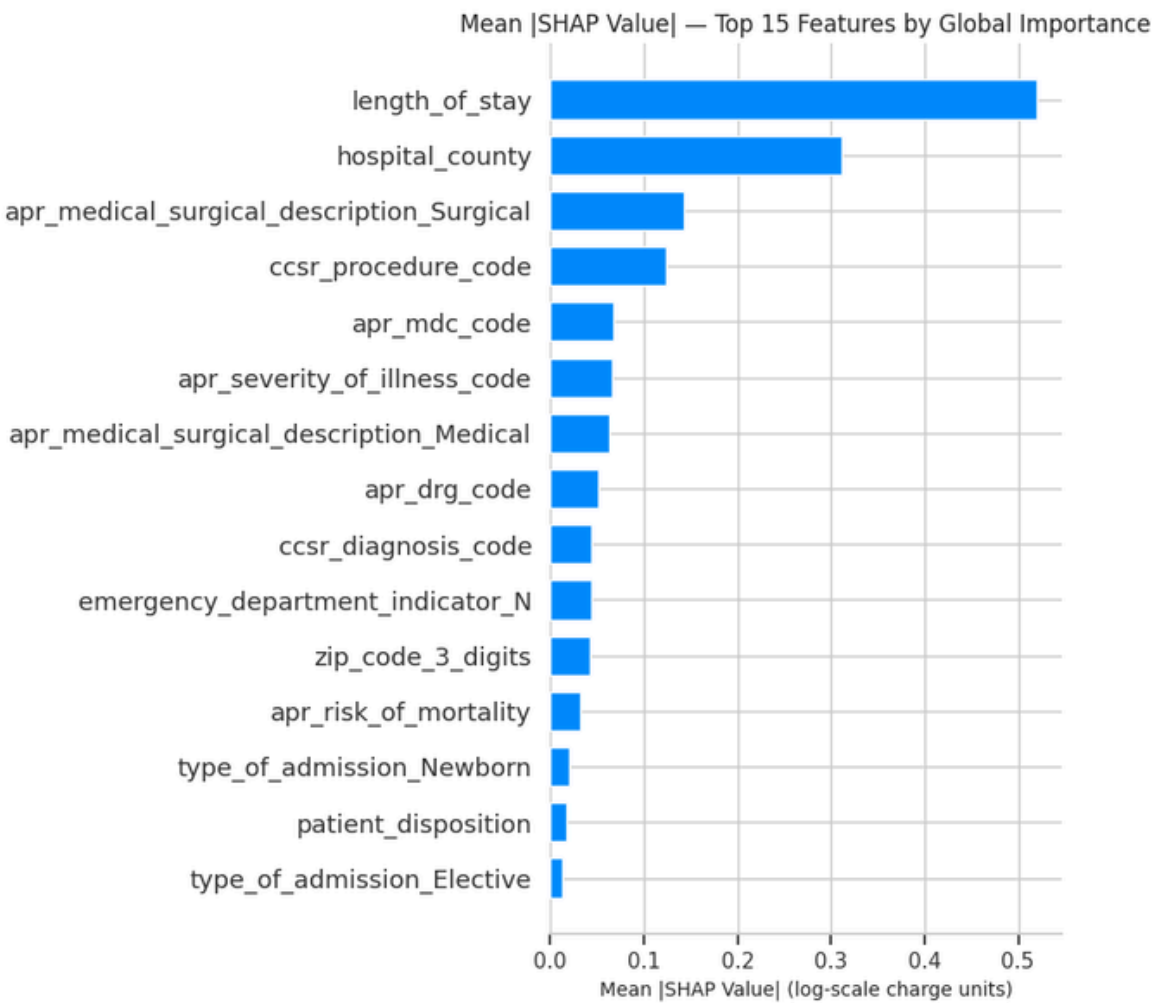
✓ With: Minor < Moderate < Major < Extreme ordering preserved; severity ranks correctly in all models

FrequencyEncoder for DRG/Diagnosis Codes

✗ Without: OHE on 500+ diagnosis codes → sparse 500-column matrix; target encoding → leakage

✓ With: Compact numerical representation; leak-free; common codes carry higher frequency signal

SHAP Analysis: length_of_stay Dominates — Severity and County Drive the Second Tier



Confirms Lasso

Top 4 SHAP features match Lasso ranking: LOS → surgical classification → county → severity.

Reveals interactions

SHAP captures nonlinear effects: LOS impact is steeper at short stays and modulated by clinical severity.

Enables audit use

Waterfall plots decompose any prediction into additive feature contributions — suitable for individual case review.

The model's $R^2=0.814$ reflects real clinical and economic signal — not noise — and the learned relationships are defensible to domain stakeholders.

Three Findings — Supported by Linear, Tree-Based, and SHAP Evidence

1

Hospital charge variation is fundamentally nonlinear

The 13-point R^2 gap between linear models (0.606) and tuned XGBoost (0.814) is not a modeling artifact — it reflects genuine interaction structure between LOS, severity, procedure type, and geography.

2

XGBoost achieves meaningful prediction accuracy

MAE=\$18,011, $R^2=0.814$. At a median charge of ~\$38,000, the model predicts within $\pm 47\%$ on average — sufficient accuracy for benchmarking and outlier flagging at the facility or diagnosis-group level.

3

Preprocessing design mattered as much as model selection

Log transformation, Winsorization, ordinal severity encoding, and frequency encoding for high-cardinality codes were each individually motivated by data characteristics — and collectively enabled the tree-based performance gains.

Thank You

Predicting Hospital Inpatient Charges — NY State SPARCS 2021

GitHub Repository

github.com/bodiase/ny-inpatient-charge-prediction

0.814

R² — tuned XGBoost

\$18,011

MAE

31%

RMSE reduction